Artificial Intelligence

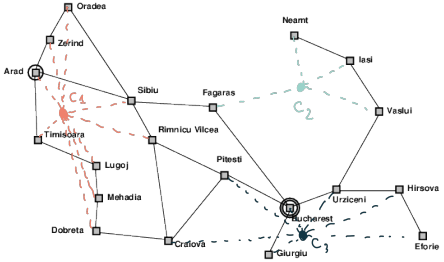
Local search in continuous space (AIMA sections 4.2)

Summary

- Intro to search in continuous space
- The concept of gradient
- Newthon Raphson
- Gradient Ascent

Example: Placing airports in Romania

Aim: locate three airports to minimize the sum of squared distances from each city to its closest airport.



- Airport loc. (x_1, y_1), (x_2, y_2), (x_3, y_3)
- City locations (x_c, y_c)
- C_a cities closest to airport a $(a = \{1, 2, 3\})$

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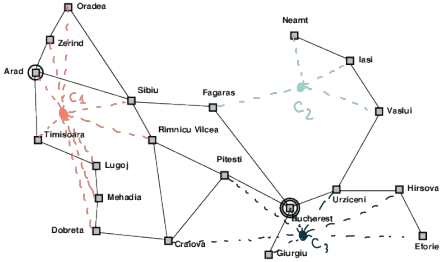


- Airport loc. (x_1, y_1), (x_2, y_2), (x_3, y_3)
- City locations (x_c, y_c)
- C_a cities closest to airport a ($a = \{1, 2, 3\}$)
- objective minimize $\sum_a \sum_{c \in C_a} (x_a - x_c)^2 + (y_a - y_c)^2$

How can we address this problem ?

Discretization methods turn continuous space into discrete space,

- fix δ
- consider $\pm\delta$ change in each coordinate
- use any of the discrete search algorithms (e.g., hill climbing)

Random perturbations

- first-choice hill climbing: keep trying until a perturbation improves the state
- simulated annealing (recall: the next state is chosen randomly)

Gradient: compute gradient of $f(X)$ analytically ($X = \{x_1, \ldots, x_n\}$)

Gradient method

- Gradient methods compute

$$\nabla f(X) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_n} \right)$$

- For the airport example: $f(X) = \sum_a \sum_{c \in C_a} (x_a - x_c)^2 + (y_a - y_c)^2$ and $X = \{x_1, y_1, x_2, y_2, x_3, y_3\}$, $n = 6$
- $\frac{\partial f}{\partial x_1} = \sum_{c \in C_1} 2(x_1 - x_c)$
- To find extremum solve for $\nabla f(X) = 0$
- Can we solve this in closed form ?

Local gradient

◇ Often we can not solve the $\nabla f(X) = 0$ equation globally, however we can improve locally:

- perform update in uphill direction for each coordinate
- the steeper the slope (the higher the derivative) the bigger the step for that coordinate

◇ Consider a general function $g(x_1, x_2)$, updates for gradient method

- $x_1 \leftarrow x_1 + \alpha * \frac{\partial g(x_1,x_2)}{\partial x_1}$, $x_2 \leftarrow x_2 + \alpha * \frac{\partial g(x_1,x_2)}{\partial x_2}$.

◇ updates, vector notation

- $X = [x_1\ x_2]^T$, $\nabla g(X) = \left[\frac{\partial g}{\partial x_1}\ \frac{\partial g}{\partial x_2}\right]^T$
- $X \leftarrow X + \alpha * \nabla g(X)$.

◇ α is the step size, a small constant, finding a proper value for α is critical

- too big: can skip solutions
- too small: too many steps required

Newton-Raphson

- Effective solution for many local search problems in continuous space
- General technique to find roots of a function, i.e., solving equation of the form

$$g(x) = 0$$

- fix an estimate of the root $\bar{x}_0$
- iteratively update the estimate: $\bar{x}_{n+1} = \bar{x}_n - \frac{g(\bar{x}_n)}{g'(\bar{x}_n)}$
- $g'(x) = dg(x)/dx$ first derivative of $g(x)$

Review: derivatives and gradient

- What is the derivative of $g(x) = x^2 + 3$, $\frac{dg}{dx} = ?$
- What is the derivative of $g(x)$ computed at $x = 5$, $\left.\frac{dg}{dx}\right|_{x=5} = ?$
- What is the gradient of $g(x, y) = x^2 y$

$$\nabla g = \begin{bmatrix} \frac{\partial g}{\partial x} \\ \frac{\partial g}{\partial y} \end{bmatrix} = \begin{bmatrix} ? \\ ? \end{bmatrix}$$

- What is the gradient of $g(x, y) = x^2 y$ computed at $x = 0.5, y = 0.5$

$$\nabla g|_{x=0.5, y=0.5} = \begin{bmatrix} ? \\ ? \end{bmatrix}$$

Exercises: Simple application of Newton method

◇ Consider the function $g(x) = x^2 - a$

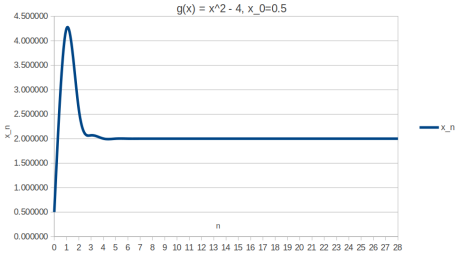
- Show that the Newton method leads to the recurrence

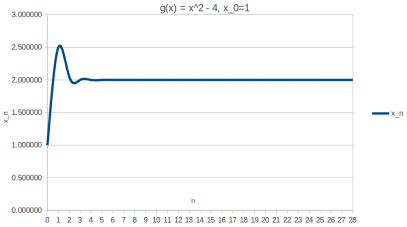
$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{a}{x_n}\right)$$

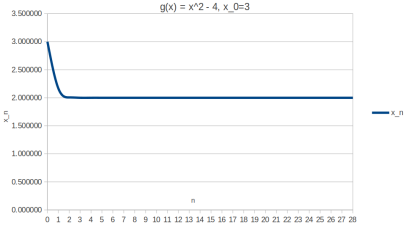
- Fix $a = 4, x_0 = 1$, compute x_i $i \in \{1, 2, 3\}$

In-dept analysis

◇ how does x_i $i \in \{1, 2, ..\}$ vary for $x_0 \in \{0.5, 1, 3\}$?



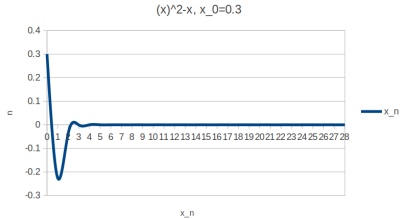


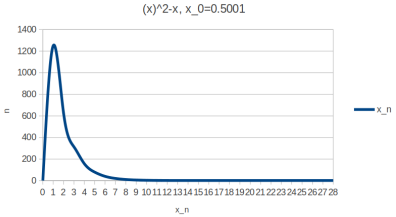


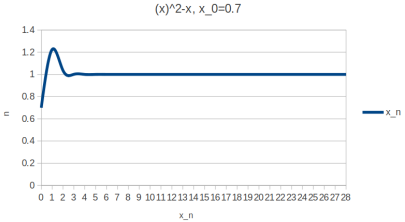
Exercise: Another simple applicaziton of Newthon method

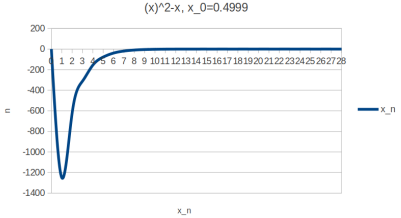
- Consider the function $g(x) = x^2 - x$
  - Compute the recurrence equation for the Newthon method
  - Fix $x_0 = 0.3$, compute x_i $i \in \{1, 2, 3\}$
  - Fix $x_0 = 0.7$, compute x_i $i \in \{1, 2, 3\}$
  - Fix $x_0 = 0.5$, compute x_i $i \in \{1, 2, 3\}$

Visualization of results









◇ x_i is not defined for $x_0 = 0.5$ but it converges to correct result for small variations.

Geometric interpretation of Newthon's method

- Assume $y = f(x)$ has a root at $x = r$;
- Tangent at the point $(a, f(a))$ is $y = f(a) + (x - a)f'(a)$
- Consider b to be the intercept for the tangent line, $0 = f(a) + (x - a)f'(a)$ then:

$$b = a - \frac{f(a)}{f'(a)}$$

- b would be the next estimate for a Newthon method starting at a
- Note: fast changes (strong non linearities) to $f(x)$ in a can result in wrong estimation of next point
  - $f(x)$ must be smooth, usually we assume that $f''(x)$ is continuous near r
- Choice of x_0 is crucial!

Back to gradient

- to find an extremum for $f(X)$ we want to find $\bar{X}$ such that $\nabla f(\bar{X}) = 0$
- hence we can use the Newthon method, where $g(X)$ is $\nabla f(X)$
- in this case the update equations in matrix-vector form becomes:
  - $\mathbf{x} \leftarrow \mathbf{x} - \mathbf{H}_f^{-1}(\mathbf{x})\nabla f(\mathbf{x})$
  - where $\mathbf{H}_{ij} = \partial^2 f / \partial x_i \partial x_j$
- for high-dimensional problems, computing the n^2 entries of the Hessian and inverting the NxN matrix may be prohibitive. Approximate versions are often used.
- RECALL this is a **local** method, hence it suffers from local maxima, plateau etc.
- Random restart and simulated annealing can be useful also in continuous spaces

Empirical Gradient

◇ Sometimes we can compute $f(X)$ for a given input but we cannot compute $\nabla f(X) = 0$ even locally

- for example, the position of a set of airport requires a large-scale simulation

◇ Empirical Gradient: response of $f(X)$ to small increment or decrement of X

◇ Perform empirical gradient search: same as hill climbing in a discretized version of the search space.